

1. Administrative & Study Identification

Full Study Title: A Study to Evaluate the Safety and Tolerability of PF-06939926 Gene Therapy in Duchenne Muscular Dystrophy **Short Title/Acronym:** Phase 1b PF-06939926 (fordadistrogene movaparvovec) in DMD **Protocol Number:** C3391001 **NCT Number:** NCT03362502 **Sponsor Name:** Pfizer **Investigational Product:** PF-06939926 (fordadistrogene movaparvovec) - Recombinant adeno-associated virus, serotype 9 (AAV9) carrying a truncated human dystrophin gene **Phase of Development:** Phase 1b **Medical Monitor:** Pfizer Clinical Operations Lead / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of a single intravenous infusion of PF-06939926 in ambulatory and non-ambulatory subjects with Duchenne muscular dystrophy (DMD). **Secondary Objectives:** To measure dystrophin expression and distribution via muscle biopsy, and to assess muscle strength, muscle quality (via MRI for muscle volume and fat fraction), and motor function.

2.2 Background & Rationale Duchenne Muscular Dystrophy (DMD) is a severe, progressive, and fatal X-linked genetic disease characterized by the absence of functional dystrophin, leading to muscle cell damage and death. There is a critical medical necessity for disease-modifying treatments. This study is conducted to test whether delivering a shortened but functional version of the dystrophin gene (mini-dystrophin) using an AAV9 viral vector can safely preserve muscle function and delay disease progression in boys with DMD.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Ascending dose) **Blinding:** Open-label **Randomization:** Non-randomized

3.2 Planned Enrollment & Duration Target \$N\$: 23 participants (enrolled) **Number of Sites:** Multi-center (United States) **Estimated Study Start:** 23-Jan-2018 **Estimated Primary Completion:** 28-Mar-2022

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male participants, aged 4 to 12 years (ambulatory and non-ambulatory). 2. Confirmed genetic diagnosis of Duchenne muscular dystrophy (DMD). 3. Receiving a stable daily dose of systemic glucocorticoids prior to enrollment.

4.2 Exclusion Criteria 1. Positive test result for neutralizing antibodies against the adeno-associated virus serotype 9 (AAV9) capsid or a positive T-cell (immune) response to dystrophin. 2. Any mutation affecting exons 9 through 13, inclusive, or a deletion affecting both exon 29 and 30. 3. Any condition or prior treatment that, in the investigator's judgment, would pose an unacceptable safety risk to the participant.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single ascending dose. Participants received one of two dose levels: low-dose (1×10^{14} vg/kg) or high-dose (3×10^{14} vg/kg). **Route of Administration:** Single Intravenous (IV) Infusion. **Duration of Treatment:** Single administration with long-term follow-up (up to 5 years).

5.2 Concomitant & Prohibited Therapy Permitted: Systemic glucocorticoids (standard of care). Up to 13 subjects were included in a cohort utilizing the concomitant immunosuppressive medication sirolimus. **Prohibited:** Other investigational gene therapies or treatments contraindicated with AAV9 delivery or systemic immunosuppression.

6. Study Timeline & Visit Schedule

Visit ID	Window	Key Activities/Procedures
Screening	Days -30 to 0	Informed Consent, AAV9 Antibody testing, T-cell response testing, Baseline MRI/Labs
Baseline/Dosing	Day 0	PF-06939926 Intravenous Infusion, Vital Signs, Safety Monitoring
Interim 1	2 Months	Safety Labs, Muscle Biopsy (Immunofluorescence for mini-dystrophin), Motor Assessments
Interim 2	Years 1 & 2	Muscle MRI (Muscle Volume and Fat Fraction), Efficacy & Motor Function Assessment
Long-Term F/U	Up to 5 Years	Final Safety Evaluation, Vitals, Long-term clinical and functional assessments

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence of Treatment-Emergent Adverse Events (TEAEs), Serious Adverse Events (SAEs), and laboratory abnormalities. **Measurement Method:** Clinical

observation, patient/caregiver reporting, and central laboratory analysis (hematology, chemistry, etc.).

7.2 Secondary & Exploratory Endpoints * Mini-dystrophin expression and distribution measured by immunofluorescence from muscle biopsies (2 months post-dosing). * Change from baseline in Muscle Volume (MV) and Fat Fraction (FF) evaluated by Magnetic Resonance Imaging (MRI). * Assessments of motor function and muscle strength.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring by the investigator; any untoward medical occurrence following the start of treatment was counted as a TEAE. Monitored specifically for risks such as myocarditis and muscle weakness. **Serious Adverse Events (SAEs):** Defined as events resulting in death, life-threatening conditions, hospitalization, or significant disability. Reported immediately to the sponsor and regulatory authorities. **Data Monitoring Committee (DMC):** An independent External Data Monitoring Committee (E-DMC) formally reviewed safety data prior to dose progression and cohort staggering to mitigate unanticipated risks.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants who received the study treatment). **Sample Size Justification:** A total of approximately 23 participants was deemed sufficient for a Phase 1b safety, tolerability, and initial signal-seeking efficacy evaluation. **Handling of Missing Data:** Due to the open-label, safety-driven nature of this Phase 1b trial, missing data was generally handled descriptively without complex multiple imputation models. Efficacy changes (e.g., MRI parameters) were analyzed by descriptive change from baseline (%CFB).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP), the Declaration of Helsinki, and applicable FDA/EMA regulations. **Monitoring Plan:** Rigorous onsite clinical monitoring to oversee gene therapy administration, immunosuppression protocols, and SAE reporting. **Audit Trail:** Standardized eCRF data entry, central laboratory integrations, and independent E-DMC safety audits.

11. Study Identifiers & Governance

Primary ID: C3391001 **Registry Number:** NCT03362502 **Sponsor/Lead Organization:** Pfizer **Study Title:** A Study to Evaluate the Safety and Tolerability of PF-06939926 Gene Therapy in Duchenne Muscular Dystrophy **Official Regulatory Title:** A PHASE 1B MULTICENTER, OPEN-LABEL, SINGLE ASCENDING DOSE STUDY TO EVALUATE THE SAFETY AND TOLERABILITY OF PF-06939926 IN AMBULATORY AND NON-AMBULATORY SUBJECTS WITH DUCHENNE MUSCULAR DYSTROPHY

12. Clinical Framework (DMD Specific)

Primary Condition: Duchenne Muscular Dystrophy (DMD) **Diagnosis Threshold:** Confirmed DMD mutation with absent functional dystrophin **Medical Methodology:** Recombinant AAV9-based Gene Therapy (fordadistrogene movaparvovec) **Optimization Target:** Successful muscle tissue transduction and stable mini-dystrophin expression

13. Study Procedures & Methodology

Management Pathway: One-time intravenous infusion of AAV9 gene vector **Education Protocol (HCP):** Site training on gene therapy handling, infusion reactions, and immunosuppression management (steroids/sirolimus) **Education Protocol (Patient):** Caregiver education on post-infusion monitoring, AE identification (e.g., signs of myocarditis), and long-term follow-up commitment **Quality Audit Mechanism:** Independent E-DMC reviews between dose escalations; central reading for muscle biopsies and MRI data

14. Observation & Follow-Up Schedule

Total Duration: Up to 5 Years follow-up **Onsite Visit Cadence:** Intensive monitoring immediately post-infusion; key onsite visits at Month 2 (Biopsy), Year 1, and Year 2 (MRI) **Remote Monitoring Frequency:** Regular safety tele-visits scheduled between major clinical milestones **Checklist Review Interval:** Ongoing monitoring for TEAEs, specifically tracking immune responses and hematologic/hepatic laboratory parameters

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				